From Local Policy Gradients to Finite-Step Direction Generation:

$A^\sharp$ Advantages for Online LLM Post-Training

Research Proposal — Restructured Working Draft

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Abstract

This proposal studies a simple question: if a reinforcement-learning subproblem admits an analytic finite-step optimum, can information about the *location of that optimum* be distilled into the optimization direction, rather than discovered only through repeated local policy-gradient updates?

We take as the primitive local RL problem the KL-proximal objective

$$\max_q \ \mathbb{E}_{\tau\sim P_q}[G(\tau)] - \eta D_{\mathrm{KL}}(P_q\|P_\pi),$$

with current/reference policy π. Ordinary policy-gradient optimization treats this as a smooth objective and follows local first-order information. In deterministic autoregressive MDPs, however, the $Q^\sharp$ identity gives the exact optimizer: the optimal trajectory distribution is an exponential tilt of the reference distribution, and its token-level conditionals can be written through a log moment-generating function of future returns. We define

$$A^\sharp_{\pi,\eta}(s,a) = \eta \log \frac{\mathbb{E}_\pi[e^{G_t/\eta} \mid s,a]}{\mathbb{E}_\pi[e^{G_t/\eta} \mid s]},$$

so that the exact finite-step target satisfies

$$q^\star_{\pi,\eta}(a \mid s) = \pi(a \mid s) e^{A^\sharp_{\pi,\eta}(s,a)/\eta}.$$

The central interpretation is that $A^\sharp$ is not merely a different advantage estimator: it is the exact finite displacement from the current policy to the proximal solution in log-policy coordinates.

The proposal is organized in three algorithmic layers. First, a *one-step fixed-reference* method directly optimizes an $A^\sharp$ surrogate whose population objective is exactly reverse-KL projection onto the analytic target. Second, the semigroup law of exact proximal operators permits a *K-step decomposition*: a hard large finite step can be split into milder subproblems with larger per-hop KL temperature, trading a longer optimization path for easier estimation. Third, we treat the proximal problem itself as a *direction generator*. Re-anchoring the reference at every current policy turns $A^\sharp$ into a finite-lookahead *secant field*, whereas ordinary natural policy gradient is the corresponding tangent field. The secant field equals an average of future tangent directions, has the same stationary points in the ideal nonparametric setting, and in a local quadratic/narrow-valley model applies the spectral transform $(I - e^{-\beta H})/\beta$, reducing stiffness and enlarging stable and non-oscillatory discretization ranges.

Because practical LLM training uses ordinary parameter gradients rather than exact natural gradients, we explicitly separate policy-space geometry from parameter-space optimization.

Under a smooth parameterization, vanilla and $A^\sharp$ gradients are the adjoint-Jacobian images of the tangent and secant signals, respectively. The exact shortest-path guarantees do not survive arbitrary neural parameterization, but finite-lookahead structure does: the $A^\sharp$ parameter gradient is the Euclidean parameter direction maximally aligned, to first order, with the finite proximal target; its expansion contains variance and higher-cumulant corrections absent from the conventional policy gradient; and under a local linear parameterization it inherits a larger stability and non-oscillation region. The proposal therefore aims to establish a precise bridge from exact finite-step control structure to practical plug-in policy optimization without claiming that ideal information-geometric results automatically transfer to neural-network SGD.

1 Introduction

Large-language-model (LLM) post-training increasingly relies on policy-gradient methods such as PPO and GRPO [1, 2]. Their practical appeal is clear: collect rollouts, estimate an advantage, and improve the actor using a first-order surrogate. Yet this standard recipe hides a basic optimization question. If the current policy is solving a *finite* policy-improvement problem, why should the optimizer use only the infinitesimal tangent information available at the current point?

This question becomes concrete in deterministic autoregressive RL. For a KL-regularized policy-improvement problem, $Q\sharp$ shows that the exact optimizer can be recovered from the current policy's future return distribution through a log moment-generating function (log-MGF) [3]. Thus, at least in this setting, the optimizer is not merely given a local slope: an analytic description of the desired finite endpoint is available. The central research question of this proposal is therefore

> *How should the location of an exact finite-step proximal solution be distilled into an iterative policy-optimization process?*

The proposal deliberately separates three levels that were mixed together in earlier versions.

1. **Foundations.** We define the KL-proximal control problem, the direct policy-gradient baseline, and the $Q\sharp$ closed-form solution in deterministic dynamics.

2. **A finite-step signal.** We define $A^\sharp$ and prove properties of the signal and its surrogate objective. These are properties of an ideal nonparametric policy-space construction, not yet of any neural optimizer.

3. **Optimization algorithms.** We study three increasingly online uses of the signal: one-step fixed-reference optimization, K-step semigroup decomposition, and a re-anchored plug-in policy-gradient method whose ideal policy-space limit is a finite-lookahead secant field.

The motivation is geometric but the intended contribution is algorithmic. Direct gradient optimization of a proximal objective initially sees the same vanilla policy gradient for every KL temperature: the desired finite endpoint changes with η, but the first derivative at the reference does not. By contrast, $A^\sharp$ exposes the exact endpoint displacement in a token-level first-order signal. This suggests a continuum between purely local policy gradient and one-shot exponential reweighting: use the finite target not necessarily as a point that must be reached in one inner solve, but as a *direction generator* that tells the actor where the locally relevant finite optimum lies.

The proposal makes several deliberately limited claims. It does not claim that higher moments are universally useful, that a distributional critic is always necessary, or that policy-space natural-gradient geometry automatically implies fewer Adam steps. In strict binary RLVR the return distribution is Bernoulli and a scalar success probability is sufficient. In arbitrary neural parameterizations, Jacobian/Fisher conditioning can distort or even hide the ideal finite-lookahead correction. The

theory instead asks which properties are exact in the ideal policy space, which survive projection into parameter space, and where approximation and estimation errors enter.

Main contributions sought.

- Derive the exact token-level finite displacement $A^\sharp$ from the closed-form KL-proximal target.
- Show that the fixed-reference $A^\sharp$ surrogate is exactly reverse-KL projection to that target, and characterize its exponential-interpolation natural-gradient path.
- Establish a target-independent log-odds shortest-path interpretation for the fixed-target $A^\sharp$ path, while explicitly restricting this result to the ideal nonparametric geometry.
- Use the proximal semigroup law to construct a K-step family interpolating between a one-shot finite target and infinitesimal natural policy gradient, and quantify the estimation–path-length–drift trade-off.
- Define a re-anchored $A^\sharp$ secant field and show that it equals a finite-horizon average of future natural-policy-gradient tangents, preserves stationary points under ideal assumptions, and improves local narrow-valley conditioning.
- Connect this ideal field to practical ordinary parameter gradients. Show that plug-in $A^\sharp$ preserves finite-target lookahead as an adjoint-Jacobian-transformed signal and yields exact first-order alignment and local stability statements without assuming access to natural gradients.
- Evaluate all three algorithmic regimes in exact deterministic trees and in LLM RLVR, with explicit measurements of target tracking, path geometry, critic error, stability, and rollout/sample efficiency.

2 Foundations: the Optimization Problem, the Local Baseline, and the Closed-Form Target

2.1 Autoregressive generation as a deterministic MDP

We consider a finite-horizon history MDP. A state s_t contains the prompt and generated prefix; an action a_t is the next token. The transition is deterministic,

$$s_{t+1} = T(s_t, a_t),$$

so all trajectory branching is controlled by the policy. Let

$$G_t = \sum_{j=t}^{H} r_j$$

denote the return from time t. Most rule-based reasoning settings are a special case with a terminal binary return $G_1 \in \{0, 1\}$, while code-test fractions and structured verifier scores provide graded deterministic rewards.

For a policy π, let $P_\pi(\tau)$ be its trajectory distribution and d^π an occupancy measure over prefix states. We assume positive support on trajectories/actions under discussion whenever log-ratios are used.

2.2 The KL-proximal objective as the primitive local RL problem

Given a reference policy π and temperature $\eta > 0$, define

$$\mathcal{F}_{\pi,\eta}(q) = \mathbb{E}_{\tau\sim P_q}[G_1(\tau)] - \eta D_{\mathrm{KL}}(P_q \| P_\pi). \tag{1}$$

Equivalently, with inverse temperature

$$\beta = \eta^{-1},$$

small β corresponds to a conservative local improvement and large β to a more aggressive finite displacement.

This proposal treats Eq. (1) as the standard *finite local policy-improvement problem*. It is not necessary that the final RL objective permanently contain a KL penalty to one initial model. In an online algorithm, the current policy may be re-used as the reference at each round, so the proximal problem can serve as a local improvement device rather than the ultimate objective.

2.3 Direct policy gradient: a local tangent method

A natural baseline is to optimize Eq. (1) directly. Let $q = \pi_\theta$ and initialize at the reference parameter θ_0 such that $\pi_{\theta_0} = \pi$. Since KL is minimized at equality,

$$\nabla_\theta D_{\mathrm{KL}}(P_{\pi_\theta} \| P_\pi)|_{\theta_0} = 0, \tag{2}$$

so

$$\boxed{\nabla_\theta \mathcal{F}_{\pi,\eta}(\pi_\theta)|_{\theta_0} = \nabla_\theta J(\pi_\theta)|_{\theta_0}, \qquad \forall \eta > 0.} \tag{3}$$

The first update therefore contains no information about how far away the intended proximal endpoint lies. Dependence on η appears only after moving away from the reference, through curvature and repeated re-linearization.

This motivates a useful geometric idealization. On the full trajectory simplex equipped with the Fisher metric, natural-gradient ascent on the unregularized expected return $J(P) = \mathbb{E}_P[G]$ has replicator dynamics

$$\dot{P}(\tau) = P(\tau)(G(\tau) - \mathbb{E}_P[G]),$$

whose flow map is the exponential-tilt semigroup

$$\Phi_u(P)(\tau) = \frac{P(\tau)e^{uG(\tau)}}{\mathbb{E}_P[e^{uG}]}. \tag{4}$$

In a deterministic autoregressive MDP this trajectory distribution is policy-realizable, and the trajectory Fisher metric pulls back to the usual occupancy-weighted conditional-policy Fisher metric. Thus ordinary nonparametric natural policy gradient can be regarded as the *tangent field* of the semigroup Φ_u.

For the frozen-reference proximal objective itself, the path can be stated explicitly.

Candidate Proposition 2.1 (Direct proximal natural-gradient flow)**.** *Let $P^\star = \Phi_\beta(P_\pi)$ be the optimizer of Eq. (1). On the trajectory simplex with the Fisher metric, natural-gradient ascent of $\mathcal{F}_{\pi,\eta}$ from $P_0 = P_\pi$ satisfies*

$$\boxed{P_t \propto P_\pi^{e^{-\eta t}} (P^\star)^{1-e^{-\eta t}} = \Phi_{\beta(1-e^{-\eta t})}(P_\pi).} \tag{5}$$

Thus direct proximal natural gradient traverses the same exponential-tilt semigroup orbit from $u = 0$ to $u = \beta$, discovering the finite endpoint through successive local tangents. Its initial direction is the ordinary NPG tangent and is independent of the chosen finite endpoint except for an overall time horizon.

This gives a clean baseline for the rest of the proposal: direct gradient optimization follows a curved semigroup orbit in conditional-policy coordinates, whereas the $A^\sharp$ construction will encode the endpoint displacement explicitly.

2.4 $Q\sharp$: the finite endpoint is available in closed form

By the Donsker–Varadhan identity, Eq. (1) has the trajectory-level optimizer

$$P_{q^\star_{\pi,\eta}}(\tau) = \frac{P_\pi(\tau)e^{G_1(\tau)/\eta}}{\mathbb{E}_\pi[e^{G_1/\eta}]} = \Phi_\beta(P_\pi)(\tau). \tag{6}$$

In a general stochastic MDP, such a trajectory tilt can reweight environment transitions and need not be realizable by changing the policy alone. Under deterministic history transitions, however, all branching is policy-controlled, so Eq. (6) factors into a valid autoregressive policy.

$Q\sharp$ identifies the corresponding token-level log-MGF quantities [3]. For a reference π define

$$Q^\sharp_{\pi,\eta}(s,a) = \eta \log \mathbb{E}_\pi[e^{G_t/\eta} \mid s_t = s, a_t = a], \tag{7}$$

$$V^\sharp_{\pi,\eta}(s) = \eta \log \mathbb{E}_\pi[e^{G_t/\eta} \mid s_t = s]. \tag{8}$$

The next section uses these objects not only to characterize a one-shot optimum, but to construct a finite-target optimization signal.

3 Motivation: Distilling a Known Finite Target into the Optimization Direction

3.1 The limitation of purely local first-order information

The distinction between conventional policy gradient and the proposed construction is not simply "mean versus distribution." The sharper distinction is *local tangent information versus finite endpoint information*.

At the reference policy, Eq. (3) says that direct first-order optimization cannot distinguish a very conservative proximal problem from an aggressive one. Geometrically, a local gradient method knows the direction that appears best at the current point, but does not know where the intended finite optimum lies. On a poorly conditioned landscape this can create a familiar pattern: successive local directions must repeatedly correct one another, producing curved paths, transverse motion, or discrete zig-zagging in stiff directions.

The availability of Eq. (6) changes the information structure. The finite proximal endpoint is analytically characterized. The central idea is therefore:

> Do not ask only for the slope of the objective at the current policy. Ask for the *relative location of the finite proximal solution*, and distill that location into a first-order signal that an actor optimizer can use.

3.2 A minimal example of first-order insufficiency

Suppose two actions have identical conditional mean return under the current policy, but different continuation-return distributions: one continuation returns 0 or 2 with probability 1/2, while the other deterministically returns 1. Conventional advantages tie because both means equal 1. Yet for finite η,

$$\eta \log \frac{1 + e^{2/\eta}}{2} > 1$$

by strict convexity. The finite proximal target therefore distinguishes the two branches even though their local mean-value advantage is identical. The distinction is not a heuristic risk bonus: it arises because future policy reweighting can amplify high-return continuations already present in the reference support.

3.3 Direction generation as the organizing idea

There are at least three ways to use the finite target.

1. **One-step fixed target.** Freeze the reference, estimate the target, and optimize the actor toward it.
2. **K-step decomposition.** Split one difficult finite target into K milder proximal subproblems, updating the reference only after each subproblem is solved.
3. **Re-anchored direction generation.** At every current policy, construct a fresh finite proximal target only to obtain its relative direction, take a small actor step, and immediately re-anchor. The proximal problem is no longer a static objective that must be solved; it becomes a local lookahead oracle.

The rest of the proposal develops these layers in this order. Before specifying any algorithm, we first isolate the exact properties of the $A^\sharp$ signal and its fixed-target surrogate.

4 $A^\sharp$ Advantage and the Fixed-Target Surrogate

4.1 $Q\sharp$ finite-step factorization, restated as a policy displacement

The following identities are not claimed as a new theorem of this proposal. They are the deterministic-$Q\sharp$ closed-form solution restated in the notation needed for our optimization viewpoint. Define

$$A^\sharp_{\pi,\eta}(s,a) = Q^\sharp_{\pi,\eta}(s,a) - V^\sharp_{\pi,\eta}(s) = \eta \log \frac{\mathbb{E}_\pi[e^{G_t/\eta} \mid s,a]}{\mathbb{E}_\pi[e^{G_t/\eta} \mid s]}. \tag{9}$$

For a finite-horizon deterministic history MDP and a full-support reference policy π, the proximal optimizer $q^\star_{\pi,\eta}$ satisfies, for every reachable (s,a),

$$\boxed{q^\star_{\pi,\eta}(a \mid s) = \pi(a \mid s) \exp\left(\frac{A^\sharp_{\pi,\eta}(s,a)}{\eta}\right).} \tag{10}$$

Equivalently,

$$\boxed{A^\sharp_{\pi,\eta}(s,a) = \eta \log \frac{q^\star_{\pi,\eta}(a \mid s)}{\pi(a \mid s)}.} \tag{11}$$

The normalization identity

$$\mathbb{E}_{a\sim\pi(\cdot|s)}\left[e^{A^\sharp_{\pi,\eta}(s,a)/\eta}\right] = 1$$

follows immediately. Our use of these identities is interpretive: Eq. (11) says that $A^\sharp/\eta$ is exactly the finite log-policy displacement from the current policy to the proximal optimum. This observation is the interface from the $Q\sharp$ solution to the actor-optimization problem studied below.

4.2 From the finite target to the $A^\sharp$ surrogate

Knowing $q^\star_{\pi,\eta}$ is not yet an actor-learning rule. We need an objective that exposes the finite displacement in Eq. (11) to an optimizer while retaining the same reverse-KL geometry relative to the reference. Fix any state weighting μ supported on states of interest and define

$$\mathcal{L}^{\mu}_{\pi,\eta}(q) = \mathbb{E}_{s\sim\mu}\left[\mathbb{E}_{a\sim q(\cdot|s)} A^\sharp_{\pi,\eta}(s,a) - \eta D_{\mathrm{KL}}(q(\cdot \mid s)\|\pi(\cdot \mid s))\right]. \tag{12}$$

Candidate Proposition 4.1 (Exact proximal-target surrogate equivalence)**.** *Using Eq.* (11)*,*

$$\boxed{\mathcal{L}^{\mu}_{\pi,\eta}(q) = -\eta \mathbb{E}_{s\sim\mu} D_{\mathrm{KL}}\big(q(\cdot \mid s)\|q^\star_{\pi,\eta}(\cdot \mid s)\big)\,.} \tag{13}$$

Thus the unrestricted maximizer is exactly $q^\star_{\pi,\eta}$ on the support of μ, while a parameterized actor computes a reverse-KL projection of the analytic target onto its representable class.

The algebra is elementary, but the identity is structurally central: it is what justifies Eq. (12) as the theory-native actor loss. It also supplies the potential function whose nonparametric Fisher natural-gradient flow is analyzed below. The equivalence holds for *any frozen state occupancy* μ, which later permits a clean separation between the policy defining the finite target and the policy used to visit states for data aggregation.

4.3 Strong-KL expansion: ordinary advantage is the infinitesimal limit

Let $\beta = 1/\eta$. The conditional cumulant expansion gives

$$Q^\sharp_{\pi,\eta}(s,a) = Q^\pi(s,a) + \frac{\beta}{2}\mathrm{Var}_\pi(G_t \mid s,a) + \frac{\beta^2}{6}\kappa_{3,\pi}(G_t \mid s,a) + O(\beta^3), \tag{14}$$

$$V^\sharp_{\pi,\eta}(s) = V^\pi(s) + \frac{\beta}{2}\mathrm{Var}_\pi(G_t \mid s) + \frac{\beta^2}{6}\kappa_{3,\pi}(G_t \mid s) + O(\beta^3). \tag{15}$$

Candidate Theorem 4.2 (Strong-KL expansion)**.** *Under bounded conditional cumulants,*

$$\boxed{A^\sharp_{\pi,\eta}(s,a) = A^\pi(s,a) + \frac{\beta}{2}\left[\mathrm{Var}(G_t \mid s,a) - \mathrm{Var}(G_t \mid s)\right] + O(\beta^2).} \tag{16}$$

Hence $A^\sharp_{\pi,\eta} \to A^\pi$ as $\eta \to \infty$ ($\beta \to 0$).

This theorem establishes consistency with ordinary policy gradient while also identifying the first finite-lookahead correction. It should not be interpreted as saying that variance is itself the objective; variance appears because the finite proximal optimum depends on the full exponential moment.

4.4 Fixed-target information geometry

For the remainder of this subsection, the reference π and target $q^\star = q^\star_{\pi,\eta}$ are frozen. We will use three state-weighting symbols deliberately, because they play different roles. Let μ_{sur} denote the particular frozen state weighting used to define the surrogate $\mathcal{L}^{\mu_{\mathrm{sur}}}_{\pi,\eta}$ in Eq. (12). The Pareto statement below may instead be evaluated under an arbitrary fixed weighting μ_{eval}; it does not have to equal μ_{sur}. By contrast, the natural-gradient-flow proposition that follows uses the μ_{sur}-weighted Fisher metric, i.e. the same weighting as the surrogate objective.

Define the exponential interpolation

$$q_\alpha(a \mid s) = \frac{\pi(a \mid s)^{1-\alpha} q^\star(a \mid s)^\alpha}{\sum_{a'} \pi(a' \mid s)^{1-\alpha} q^\star(a' \mid s)^\alpha}, \qquad \alpha \in [0, 1]. \tag{17}$$

By Eq. (11),

$$q_\alpha(a \mid s) \propto \pi(a \mid s) \exp\left(\frac{\alpha A^\sharp_{\pi,\eta}(s, a)}{\eta}\right).$$

Importantly, the pointwise path q_α itself does not depend on any state weighting.

Candidate Theorem 4.3 (Target-KL Pareto optimality of the exponential path)**.** *Fix* any *state weighting* μ_{eval} *on the states of interest and define*

$$D_{\mu_{\text{eval}}}(q\|r) = \mathbb{E}_{s\sim\mu_{\text{eval}}} D_{\text{KL}}(q(\cdot \mid s)\|r(\cdot \mid s)).$$

Let

$$R_{\text{eval}}(\alpha) = D_{\mu_{\text{eval}}}(q_\alpha\|\pi), \qquad T_{\text{eval}}(\alpha) = D_{\mu_{\text{eval}}}(q_\alpha\|q^\star).$$

For every feasible radius

$$0 \le \epsilon \le D_{\mu_{\text{eval}}}(q^\star\|\pi),$$

there exists $\alpha(\epsilon)$ *with* $R_{\text{eval}}(\alpha(\epsilon)) = \epsilon$*, and*

$$\boxed{q_{\alpha(\epsilon)} = \arg\min_q D_{\mu_{\text{eval}}}(q\|q^\star) \quad s.t. \quad D_{\mu_{\text{eval}}}(q\|\pi) \le \epsilon.} \tag{18}$$

Under nondegeneracy the solution is unique on the support of μ_{eval}*.*

Thus the same pointwise exponential path is Pareto-optimal for every fixed choice of μ_{eval}, provided that the *same* μ_{eval} is used in both the radius constraint and the distance-to-target objective. Choosing $\mu_{\text{eval}} = \mu_{\text{sur}}$ makes the theorem directly describe the geometry of the surrogate in Eq. (12), but this equality is not mathematically required for the Pareto statement itself.

Candidate Proposition 4.4 (Natural-gradient flow of the frozen surrogate)**.** *Equip the conditional policy simplex with the* μ_{sur}*-weighted Fisher metric, using the* same μ_{sur} *that defines* $\mathcal{L}^{\mu_{\text{sur}}}_{\pi,\eta}$*. The natural-gradient ascent flow of this frozen surrogate, initialized at* $q_0 = \pi$*, remains on Eq. (17):*

$$q_t = q_{\alpha(t)}, \qquad \alpha(t) = 1 - e^{-\eta t}. \tag{19}$$

Equivalently, all pairwise log-odds errors to $q^\star$ *contract exponentially at the same rate* η*.*

The matching of state weightings in Proposition 4.4 is substantive, not cosmetic. If the surrogate uses μ_{sur} but the Fisher metric uses another equivalent weighting μ_{met}, then at state s the pairwise log-odds error contracts at rate $\eta\, d\mu_{\text{sur}}/d\mu_{\text{met}}(s)$. Each state still moves along its own exponential segment, but a common scalar $\alpha(t)$ across all states generally disappears unless the two weightings are proportional on their common support.

4.5 A target-independent log-odds shortest-path view

The Pareto theorem above measures progress by the reverse-KL distance to the distinguished target $q^\star$. This is exactly the right criterion for target tracking, but it does not by itself show that the path is short under an *ambient policy geometry whose definition does not single out* $q^\star$. To obtain such a complementary statement, fix any state weighting $\mu_{\log}$ and full-support action measure $\nu(\cdot \mid s)$, independently of the target, and define centered log-policy coordinates

$$z_\pi^{(\nu)}(s,a) = \log \pi(a \mid s) - \mathbb{E}_{b\sim\nu(\cdot|s)} \log \pi(b \mid s).$$

Set

$$d^2_{\log,\mu_{\log},\nu}(\pi,\rho) = \mathbb{E}_{s\sim\mu_{\log},a\sim\nu}\left[z_\pi^{(\nu)}(s,a) - z_\rho^{(\nu)}(s,a)\right]^2. \tag{20}$$

Equivalently,

$$d^2_{\log,\mu_{\log},\nu}(\pi,\rho) = \frac{1}{2}\mathbb{E}_{s\sim\mu_{\log},a,b\sim\nu}\left[\log \frac{\pi(a \mid s)\rho(b \mid s)}{\pi(b \mid s)\rho(a \mid s)}\right]^2,$$

so the metric measures root-mean-square changes in pairwise action log-odds.

Candidate Theorem 4.5 (Universal log-odds shortest path for a fixed finite target)**.** *For every fixed full-support pair* $(\mu_{\log},\nu)$*, the path* q_α *in Eq. (17) is affine in* $z^{(\nu)}$*:*

$$z_{q_\alpha}^{(\nu)} = (1-\alpha)z_\pi^{(\nu)} + \alpha z_{q^\star}^{(\nu)}.$$

Hence among all absolutely continuous policy paths with the same endpoints,

$$L_{\mu_{\log},\nu}[q] = \int_0^1 \|\dot z_t\|_{\mu_{\log},\nu}\, dt \geq d_{\log,\mu_{\log},\nu}(\pi,q^\star),$$

with equality for the $A^\sharp$ *path. Any genuine transverse excursion seen by* $(\mu_{\log},\nu)$ *yields strict excess length.*

The direct proximal/NPG path generally does not share this affine property. If

$$M_{s,a}(u) = \mathbb{E}_\pi[e^{uG_t} \mid s,a],$$

then the direct semigroup path has conditional log weights $\log M_{s,a}(\alpha\beta)$, whereas Eq. (17) uses $\alpha \log M_{s,a}(\beta)$. Its log-odds curvature is generated by conditional tilted variances,

$$\frac{d^2}{d\alpha^2}\log M_{s,a}(\alpha\beta) = \beta^2 \mathrm{Var}_{\text{tilted at }\alpha\beta}(G_t \mid s,a).$$

This formalizes the narrow-valley intuition for a *fixed target*: $A^\sharp$ pre-integrates finite endpoint information into a straight log-odds displacement. Again, this theorem concerns the ideal nonparametric policy path, not arbitrary parameter-space SGD.

5 Algorithm I: One-Step $A^\sharp$ Surrogate Optimization

5.1 Direct parameterized optimization of the fixed surrogate

The most direct algorithmic use of Section 4 is simply to parameterize the candidate policy in Eq. (12). Fix a reference policy π, a finite temperature η, a frozen state weighting μ, and the corresponding quantities

$$A^\sharp = A^\sharp_{\pi,\eta}, \qquad q^\star = q^\star_{\pi,\eta}.$$

Let π_θ be a parameterized actor, initialized from the reference when appropriate. The one-step algorithm optimizes

$$\mathcal{L}(\theta) = \mathbb{E}_{s\sim\mu}\left[\mathbb{E}_{a\sim\pi_\theta(\cdot|s)}A^\sharp(s,a) - \eta D_{\mathrm{KL}}(\pi_\theta(\cdot \mid s)\|\pi(\cdot \mid s))\right]. \tag{21}$$

By Proposition 4.1, this is exactly

$$\boxed{\mathcal{L}(\theta) = -\eta D_\mu(\pi_\theta\|q^\star).} \tag{22}$$

Here "one-step" means that there is one fixed proximal problem and one fixed target $q^\star$; it does *not* mean one gradient update, one rollout round, or an outer loop.

The distinction from Section 4 is essential. The actor is updated with the *ordinary parameter gradient*

$$\theta \leftarrow \theta + \gamma\nabla_\theta\mathcal{L}(\theta),$$

not the Fisher natural gradient on the nonparametric policy simplex. Consequently, the exponential interpolation, KL-budget Pareto path, and log-odds shortest-path statements of Section 4 do not generally describe the trajectory taken by parameter-space SGD or Adam. What remains exact is the objective itself: every ordinary gradient step is a gradient step on the parameterized reverse-KL objective in Eq. (22), and any global optimum reached within the actor class is a reverse-KL projection of $q^\star$ onto that class. No stronger path-optimality claim is made for this algorithm.

5.2 Sampling form and the distinction between reference and behavior policies

The reference policy and behavior policy play different roles even though there is no segmentation yet.

- The **reference policy** π defines the proximal problem, the continuation-return distribution, $A^\sharp$, and the fixed target $q^\star$. It remains fixed throughout this one-target optimization problem.
- A **behavior policy** b determines which states and actions are sampled for estimating the fixed surrogate. It may coincide with π, but conceptually it is a sampling object rather than part of the target definition.

For samples from a fixed behavior policy b, define

$$w_\theta(s,a) = \frac{\pi_\theta(a \mid s)}{b(a \mid s)}.$$

At a fixed state weighting μ, an unbiased importance-sampling form of Eq. (21) is

$$\mathcal{L}(\theta) = \mathbb{E}_{s\sim\mu,\, a\sim b(\cdot|s)}\left[w_\theta(s,a)\left(A^\sharp(s,a) - \eta\log\frac{\pi_\theta(a \mid s)}{\pi(a \mid s)}\right)\right]. \tag{23}$$

In the clean on-reference case $b = \pi$ and at initialization $\pi_{\theta_0} = \pi$, the KL gradient vanishes and

$$\nabla_\theta\mathcal{L}(\theta_0) = \mathbb{E}_{s\sim\mu,\, a\sim\pi}\left[\nabla_\theta\log\pi_{\theta_0}(a \mid s)A^\sharp(s,a)\right]. \tag{24}$$

Thus the first ordinary parameter-gradient direction depends explicitly on the finite target through η, unlike direct first-order differentiation of the original proximal objective at its reference.

Because Proposition 4.1 holds for any frozen μ, state coverage need not equal the reference occupancy. In principle one may roll in with an updated behavior policy to visit more relevant

prefixes while still evaluating continuation labels under the frozen reference policy. This changes the sampling distribution without changing the supervised target. By contrast, if updated actors are also used to generate the continuations that define $A^\sharp$, then the reference-dependent target itself changes; that is no longer the fixed one-step problem and belongs to the re-anchored construction of Section 7.

5.3 Why the one-step algorithm is statistically difficult

The attractive geometry of the fixed target becomes hardest exactly when the finite displacement is largest. For small η (large β), $e^{\beta G}$ concentrates on rare high-return events, so estimating

$$h_\pi(s,a) = \mathbb{E}_\pi[e^{\beta G_t} \mid s,a]$$

becomes statistically demanding. If

$$\widehat{A}^\sharp = A^\sharp + \varepsilon, \qquad \widehat{q}(a \mid s) \propto \pi(a \mid s) e^{\widehat{A}^\sharp(s,a)/\eta},$$

then for small centered error,

$$D_{\mathrm{KL}}(\widehat{q} \| q^\star) = \frac{\mathrm{Var}_{q^\star}[\varepsilon]}{2\eta^2} + O(\|\varepsilon\|^3 / \eta^3). \tag{25}$$

Thus aggressive one-shot targeting trades approximation bias for estimation sensitivity.

5.4 Practical actor variants

The theory-native actor objective is Eq. (21). Two useful approximations are:

- **Forward-KL distillation:** explicitly form $q^\star$ and minimize $D_\mu(q^\star \| \pi_\theta)$. This shares the unrestricted optimum but has different finite-capacity projection bias.
- **PPO-style clipping:** replace the explicit reverse-KL objective by a clipped importance-ratio objective using $A^\sharp$. This is easy to integrate into PPO/GRPO codebases but breaks the exact reverse-KL projection identity and must be presented as a practical approximation rather than as the object analyzed in Section 4.

6 Algorithm II: K-Step $A^\sharp$ Surrogate Optimization

6.1 The proximal semigroup law

Let $\mathcal{T}_\beta$ denote the exact trajectory-level tilt operator

$$P_{\mathcal{T}_\beta(\pi)}(\tau) \propto P_\pi(\tau) e^{\beta G(\tau)}.$$

Candidate Proposition 6.1 (Semigroup composition of exact proximal steps)**.** *For $\beta_1, \beta_2 \geq 0$,*

$$\boxed{\mathcal{T}_{\beta_2} \circ \mathcal{T}_{\beta_1} = \mathcal{T}_{\beta_1+\beta_2}.} \tag{26}$$

Equivalently, in temperature notation,

$$\frac{1}{\eta_{\mathrm{eff}}} = \frac{1}{\eta_1} + \frac{1}{\eta_2}.$$

Suppose the desired one-shot problem has $(\beta_\star, \eta_\star)$ with $\beta_\star = 1/\eta_\star$. Splitting it into K equal exact hops uses

$$\beta_{\mathrm{hop}} = \frac{\beta_\star}{K}, \qquad \eta_{\mathrm{hop}} = K\eta_\star. \tag{27}$$

If every inner problem is solved exactly, the final endpoint is unchanged:

$$\mathcal{T}^K_{\beta_\star/K}(\pi_0) = \mathcal{T}_{\beta_\star}(\pi_0).$$

6.2 Why splitting can help despite identical exact endpoints

The value of splitting is not a better ideal endpoint. It is a trade-off among estimation difficulty, optimization path, and reference refresh.

Each hop has a larger temperature and therefore a milder exponential transform. By Theorem 4.2,

$$A^\sharp_{\pi,\eta_{\mathrm{hop}}} = A^\pi + O(K^{-1}),$$

for a fixed total $\beta_\star$. Thus increasing K makes each critic target easier and less tail-sensitive.

The price is geometric. The $K = 1$ fixed-target natural-gradient path is the straight exponential interpolation between π_0 and $\mathcal{T}_{\beta_\star}(\pi_0)$ in log-odds coordinates. With $K > 1$, the reference is refreshed at intermediate exact points

$$\pi^\star_j = \mathcal{T}_{j\beta_\star/K}(\pi_0),$$

and each inner solve follows the local straight chord from $\pi^\star_{j-1}$ to $\pi^\star_j$. The concatenation is generally not the same as the global one-shot straight path.

Candidate Proposition 6.2 (Path-length cost and precise $K \to \infty$ limit)**.** *Fix a full-support log-odds Hilbert geometry $(\mu_{\log}, \nu)$ and write*

$$z(u) = z^{(\nu)}_{\mathcal{T}_u(\pi_0)}, \qquad 0 \le u \le \beta_\star.$$

Let $\Delta u = \beta_\star/K$, $u_j = j\Delta u$, and let $\widetilde{z}_K(u)$ be the piecewise-affine interpolation of the exact hop endpoints $z(u_j)$. Because each frozen-target inner natural-gradient path is an exponential interpolation, $\widetilde{z}_K$ is exactly the log-odds image of the geometrically parameterized piecewise K-hop path.

First, for every finite K, its total length satisfies

$$L_{\mu_{\log},\nu}[\widetilde{z}_K] \ge \|z(\beta_\star) - z(0)\|_{\mu_{\log},\nu} = d_{\log,\mu_{\log},\nu}\left(\pi_0, \mathcal{T}_{\beta_\star}(\pi_0)\right).$$

Second, suppose $z : [0, \beta_\star] \to \mathcal{H}_{\mu_{\log},\nu}$ is C^2 in the corresponding log-odds Hilbert space. Then, as $K \to \infty$,

$$\boxed{\sup_{u\in[0,\beta_\star]} \|\widetilde{z}_K(u) - z(u)\|_{\mu_{\log},\nu} = O(K^{-2}),} \tag{28}$$

and, for the a.e. derivative of the piecewise-affine path,

$$\boxed{\operatorname*{ess\,sup}_{u\in[0,\beta_\star]} \left\|\dot{\widetilde{z}}_K(u) - \dot{z}(u)\right\|_{\mu_{\log},\nu} = O(K^{-1}).} \tag{29}$$

Since the semigroup orbit satisfies

$$\dot{z}(u) = X_{\mathrm{NPG}}(z(u)),$$

Eqs. (28)–(29) make the phrase "$K \to \infty$ approaches local NPG" precise: the piecewise finite-target path converges uniformly to the nonparametric NPG orbit, while its chord velocities converge essentially uniformly to the NPG tangent field along that orbit.

Thus the continuum should be read geometrically, not as a statement about wall-clock optimizer time:

$$\boxed{\begin{array}{c} K=1:\ \text{one global finite-target chord} \\ \downarrow \\ 1<K<\infty:\ \text{piecewise finite-target chords} \\ \downarrow \\ K\to\infty:\ \widetilde{z}_K\to z_{\mathrm{NPG}},\quad \dot{\widetilde{z}}_K\to X_{\mathrm{NPG}}(z_{\mathrm{NPG}}) \end{array}} \tag{30}$$

Thus K is not merely an engineering parameter. It trades finite-target path efficiency against the statistical ease of estimating each hop.

The same source of detour has a parameterized analogue, although the exact path-length theorem no longer survives arbitrary neural parameterization. In the one-step algorithm, ordinary parameter optimization keeps descending a *single* fixed projection objective

$$D_{\mu_{\mathrm{sur}}}(\pi_\theta\|q_K^\star), \qquad q_K^\star=\mathcal{T}_{\beta_\star}(\pi_0).$$

In a K-step implementation, this is replaced by a sequence of objectives

$$D_{\mu_j}(\pi_\theta\|q_j^\star), \qquad q_j^\star=\mathcal{T}_{j\Delta u}(\pi_0),$$

so the Euclidean parameter gradient is repeatedly reoriented toward changing intermediate targets. This can create additional parameter-space motion even though the exact policy-space endpoints obey the semigroup law. No unconditional neural-network path-length ordering follows, because the map $\theta\mapsto\pi_\theta$, the optimizer, and the state weightings can all distort geometry. Under a fixed local linearization $z_{\pi_\theta}\approx z_0+B(\theta-\theta_0)$ with common weighting and exact local representability, however, the corresponding minimum-norm parameter displacements obey the polygonal bound

$$\sum_{j=1}^{K}\left\|B^+\big(z(u_j)-z(u_{j-1})\big)\right\|_2\geq\left\|B^+\big(z(\beta_\star)-z(0)\big)\right\|_2,$$

which is the local parameter-space analogue of the policy-space detour picture. This bound concerns target displacements, not the exact trajectory taken by SGD/Adam.

6.3 Inexact inner solves and accumulated drift

The exact semigroup law requires each reference update to use the exact previous optimum. If inner optimization is stopped early, the next proximal problem is anchored at the wrong policy and errors can accumulate.

A simple idealized bound can be stated in trajectory space using the Hilbert projective metric

$$d_H(P,Q)=\operatorname{ess\,sup}\log\frac{P}{Q}-\operatorname{ess\,inf}\log\frac{P}{Q}.$$

Exponential tilting by the same reward is an isometry in this metric.

Candidate Proposition 6.3 (Additive accumulation of inexact-hop error)**.** *Suppose an approximate K-step trajectory sequence satisfies*

$$d_H(\widehat{P}_{j+1},\mathcal{T}_{\beta_j}(\widehat{P}_j))\leq\varepsilon_j.$$

Then

$$\boxed{d_H\left(\widehat{P}_K,\mathcal{T}_{\sum_j\beta_j}(P_0)\right)\leq\sum_{j=0}^{K-1}\varepsilon_j.} \tag{31}$$

The bound is intentionally idealized, but it formalizes the core concern: a staged method can make each estimation problem easier while introducing repeated projection/optimization error. This motivates an alternative that does not require any inner subproblem to converge before the reference is refreshed.

7 Algorithm III: Re-Anchored $A^\sharp$ as a Plug-In Direction Generator

7.1 From a static proximal objective to a local direction oracle

Fix a lookahead inverse temperature $\beta > 0$. At an arbitrary current policy π, define a fresh finite proximal target

$$\mathcal{T}_\beta(\pi) = \arg\max_q \left\{ J(q) - \beta^{-1} D_{\mathrm{KL}}(P_q \| P_\pi) \right\},$$

and, for brevity, write $A^\sharp_{\pi,\beta} \equiv A^\sharp_{\pi,\eta=1/\beta}$ for the corresponding finite-step advantage. We do *not* require the actor to converge to $\mathcal{T}_\beta(\pi)$. Instead, the target is used only to specify where the actor should move next.

This viewpoint is especially natural for an online PPO-like algorithm: roll out from π_k, estimate $A^\sharp_{\pi_k,\beta}$, replace the conventional advantage by this signal for one or a few small actor updates, set the new actor as the next reference, and repeat.

7.2 Tangent field versus secant field

Let $z(\pi)$ denote any fixed centered log-policy coordinate or the equivalent collection of pairwise log-odds. Let $\Phi_u = \mathcal{T}_u$ be the exact natural-policy-gradient flow map of expected reward in the ideal deterministic nonparametric setting. Define the ordinary tangent field in these coordinates by

$$X(z(\pi)) = \frac{d}{du} z(\Phi_u(\pi))\Big|_{u=0}.$$

The re-anchored $A^\sharp$ field is

$$\boxed{Y_\beta(z(\pi)) = \frac{z(\Phi_\beta(\pi)) - z(\pi)}{\beta}.} \tag{32}$$

For each pair of actions,

$$Y_{\beta,ab}(s) = A^\sharp_{\pi,\beta}(s,a) - A^\sharp_{\pi,\beta}(s,b).$$

Candidate Theorem 7.1 (Secant representation and future-tangent averaging)**.** *Along the ordinary NPG semigroup,*

$$\boxed{Y_\beta(z) = \frac{1}{\beta} \int_0^\beta X(\Phi_u(z))\, du.} \tag{33}$$

If X is differentiable,

$$Y_\beta(z) = X(z) + \frac{\beta}{2} DX(z)[X(z)] + O(\beta^2). \tag{34}$$

Thus ordinary NPG uses the instantaneous tangent, while the re-anchored $A^\sharp$ field uses the chord to a point β units ahead along the exact NPG orbit. Equation (33) makes the lookahead interpretation precise: the secant direction averages the local directions that ordinary NPG would discover only after moving forward and repeatedly re-linearizing.

7.3 Ideal policy-space properties of the secant field

Candidate Theorem 7.2 (Stationary-point preservation)**.** *Assume $X = \mathrm{grad}_h J$ is a natural-gradient field for which J strictly increases along every nonstationary trajectory. Then for every $\beta > 0$,*

$$\boxed{Y_\beta(z) = 0 \quad \Longleftrightarrow \quad X(z) = 0.} \tag{35}$$

Indeed, $Y_\beta(z) = 0$ implies $\Phi_\beta(z) = z$. But

$$J(\Phi_\beta(z)) - J(z) = \int_0^\beta \|\mathrm{grad}_h J(\Phi_u(z))\|_h^2 \, du,$$

so equality requires a stationary trajectory. Thus finite lookahead changes dynamics without inventing new stationary points in this ideal setting.

Candidate Theorem 7.3 (Convergence under a contractive NPG flow map)**.** *Suppose, in a Hilbert log-policy coordinate norm, the finite NPG flow map satisfies*

$$\|\Phi_\beta(z) - z^\star\| \le q_\beta \, \|z - z^\star\| \,, \qquad q_\beta < 1.$$

Then the secant ODE

$$\dot{z} = Y_\beta(z)$$

converges exponentially to the same fixed point:

$$\|z_t - z^\star\| \le \exp\!\left(-\frac{1 - q_\beta}{\beta} t\right) \|z_0 - z^\star\| \,.$$

This is a sufficient, not universal, global convergence condition. The more informative structural result arises from a local narrow-valley model.

7.4 Local narrow-valley analysis: finite lookahead as spectral damping

Assume that near an optimum $z^\star$ the ordinary tangent field linearizes as

$$\dot{e} = -He, \qquad e = z - z^\star, \qquad H = H^\top \succ 0. \tag{36}$$

Then

$$\Phi_\beta(e) = e^{-\beta H} e.$$

Hence the re-anchored secant field is

$$\boxed{\dot{e} = -H_\beta e, \qquad H_\beta = \frac{I - e^{-\beta H}}{\beta}.} \tag{37}$$

Candidate Theorem 7.4 (Spectral conditioning of the secant field)**.** *If $Hv_i = \lambda_i v_i$, then*

$$H_\beta v_i = \phi_\beta(\lambda_i) v_i, \qquad \phi_\beta(\lambda) = \frac{1 - e^{-\beta\lambda}}{\beta}.$$

For $\lambda > 0$, $\phi_\beta(\lambda) < \lambda$, and $\phi_\beta(\lambda)/\lambda$ decreases with λ. Therefore, if $\lambda_{\max} > \lambda_{\min} > 0$,

$$\boxed{\kappa(H_\beta) = \frac{1 - e^{-\beta\lambda_{\max}}}{1 - e^{-\beta\lambda_{\min}}} < \frac{\lambda_{\max}}{\lambda_{\min}} = \kappa(H).} \tag{38}$$

Small-curvature directions satisfy $\phi_\beta(\lambda) = \lambda + O(\beta\lambda^2)$, while stiff directions saturate at approximately $1/\beta$. Finite lookahead therefore acts as a spectral compression of the local tangent field.

For the continuous linear flow, the arc length obeys

$$\|e_0\| \le L_\beta \le \kappa(H_\beta)\,\|e_0\|\,.$$

Within the linear model, $H_\beta \to I/\beta$ as $\beta \to \infty$, so the trajectory approaches the radial straight line to the optimum and $L_\beta \to \|e_0\|$. This is a local model result, not a global statement that every nonlinear secant trajectory is shorter than every tangent trajectory.

7.5 Discrete policy-space integration: stability and oscillation

Euler discretization of the tangent and secant linear systems gives

$$e^{\tan}_{n+1} = (I - \gamma H)e_n, \tag{39}$$
$$e^{\sec}_{n+1} = (I - \gamma H_\beta)e_n. \tag{40}$$

The tangent iteration is stable for $0 < \gamma < 2/\lambda_{\max}(H)$, while the secant iteration is stable for

$$0 < \gamma < \frac{2}{\lambda_{\max}(H_\beta)},$$

which is a strictly larger interval. Moreover, a tangent eigenmode flips sign when $\gamma\lambda_i > 1$. Since $\lambda_{\max}(H_\beta) < \lambda_{\max}(H)$, there exists a nonempty step-size interval in which the stiffest tangent mode oscillates while no secant mode flips sign. This is the precise local analogue of narrow-valley zig-zag suppression.

7.6 Why this ideal field is relevant to a plug-in online algorithm

Consider the unclipped PPO-like surrogate

$$\mathcal{L}^{\mathrm{plug}}_k(\theta) = \mathbb{E}_{\tau\sim\pi_k} \sum_t \rho_t(\theta)\, \widehat{A}^{\sharp}_{\pi_k,\beta,t}, \qquad \rho_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_k(a_t \mid s_t)}. \tag{41}$$

At $\theta = \theta_k$,

$$\nabla_\theta \mathcal{L}^{\mathrm{plug}}_k(\theta_k) = \mathbb{E}_{\pi_k} \sum_t \nabla_\theta \log \pi_k(a_t \mid s_t) \widehat{A}^{\sharp}_{\pi_k,\beta,t}. \tag{42}$$

With exact $A^\sharp$, fresh re-anchoring, and an infinitesimal natural/mirror policy step, Eq. (42) integrates the secant field above. The practical problem is that neural LLM training uses an ordinary Euclidean parameter gradient, not that natural policy step. The next subsection analyzes exactly what survives.

7.7 Gradient-flow correspondence and what ordinary parameter gradients preserve

Let

$$\Phi : \Theta \to \mathcal{P}, \qquad \Phi(\theta) = \pi_\theta$$

be a smooth policy parameterization.

Candidate Proposition 7.5 (Necessary and sufficient condition for policy-gradient-flow equivalence)**.** *Suppose a discrete optimizer has*

$$\theta_{n+1} = \theta_n + \gamma V(\theta_n) + o(\gamma),$$

so its continuous limit is $\dot{\theta} = V(\theta)$*. It induces a prescribed policy-manifold gradient flow* $\dot{\pi} = \operatorname{grad}_h \mathcal{J}(\pi)$ *if and only if*

$$D\Phi_\theta[V(\theta)] = \operatorname{grad}_h \mathcal{J}(\pi_\theta)$$

along the trajectory. If the parameter dynamics itself is a Riemannian gradient flow with metric g*, the additional equivalent requirement is*

$$V(\theta) = \operatorname{grad}_g(\mathcal{J} \circ \Phi)(\theta).$$

Ordinary SGD/Adam generally does *not* satisfy this condition for the Fisher natural-gradient secant field. Therefore the ideal shortest-path and condition-number results above must not be claimed verbatim for a neural actor. Nevertheless, the lookahead signal has a precise parameter-space image.

For frozen μ define the score operator

$$(\mathcal{B}_\theta v)(s,a) = \langle \nabla_\theta \log \pi_\theta(a \mid s), v \rangle ,$$

with $L^2(\mu \otimes \pi)$ inner product, and adjoint

$$\mathcal{B}_\theta^* f = \mathbb{E}_{s\sim\mu, a\sim\pi}[\nabla_\theta \log \pi_\theta(a \mid s) f(s,a)].$$

Then $F_\theta = \mathcal{B}_\theta^* \mathcal{B}_\theta$ is the corresponding Fisher matrix.

Let X_π be the centered conventional-advantage tangent signal and $Y_{\beta,\pi}$ the centered $A^\sharp$ secant signal. Their ordinary parameter gradients are

$$\boxed{g_{\mathrm{PG}}(\theta) = \mathcal{B}_\theta^* X_\pi, \qquad g_\beta^\sharp(\theta) = \mathcal{B}_\theta^* Y_{\beta,\pi}.} \tag{43}$$

Thus replacing A^π by $A^\sharp$ does not preserve the policy-space secant exactly; it passes both tangent and secant signals through the same parameterization-dependent adjoint Jacobian.

7.8 Finite-target alignment with an ordinary Euclidean parameter step

Define the exact finite log-policy displacement

$$\Delta_\beta(s,a) = \log \frac{\mathcal{T}_\beta(\pi)(a \mid s)}{\pi(a \mid s)} = \beta A^\sharp_{\pi,\beta}(s,a).$$

For a small parameter step v,

$$\log \pi_{\theta+v} - \log \pi_\theta = \mathcal{B}_\theta v + O(\|v\|^2).$$

Candidate Theorem 7.6 (Finite-target first-order alignment under Euclidean parameter budgets)**.** *Among all parameter directions with* $\|v\|_2 \le c$*, the direction maximizing the first-order alignment of the induced log-policy displacement with the finite proximal target,*

$$\max_{\|v\|_2 \le c} \langle \mathcal{B}_\theta v, \Delta_\beta \rangle_{\mu,\pi} ,$$

is

$$\boxed{v^\star = c \frac{\mathcal{B}_\theta^* \Delta_\beta}{\|\mathcal{B}_\theta^* \Delta_\beta\|} \propto g_\beta^\sharp(\theta).} \tag{44}$$

This is a direct ordinary-gradient guarantee: even without F^{-1}, the plug-in $A^\sharp$ gradient is the Euclidean parameter direction best aligned, to first order, with the exact finite target. Natural gradient would additionally compensate for the parameterization metric; ordinary gradient retains the target information but not the invariant geometry.

The cumulant expansion also survives projection:

$$\boxed{g_\beta^\sharp = g_{\mathrm{PG}} + \frac{\beta}{2}\mathbb{E}[\nabla_\theta \log \pi \,\mathrm{Var}(G_t \mid s,a)] + \frac{\beta^2}{6}\mathbb{E}[\nabla_\theta \log \pi \,\kappa_3(G_t \mid s,a)] + O(\beta^3),} \tag{45}$$

where state-only baselines vanish under the score expectation. Thus finite-lookahead corrections remain present in the ordinary parameter gradient.

7.9 Local ordinary-gradient stability under parameterization

A weaker but still useful analogue of the policy-space narrow-valley theorem can be obtained locally. Work in locally whitened log-policy coordinates for the fixed policy-space inner product, and suppose

$$z - z^\star \approx B(\theta - \theta^\star)$$

with locally constant full-column-rank Jacobian B, and suppose the ideal tangent linearization is Eq. (36). Then the corresponding local Euclidean parameter-gradient error dynamics have matrices

$$K = B^\top H B, \qquad K_\beta = B^\top H_\beta B. \tag{46}$$

Since $0 \prec H_\beta \prec H$,

$$\boxed{0 \prec K_\beta \prec K, \qquad \lambda_{\max}(K_\beta) < \lambda_{\max}(K).} \tag{47}$$

Therefore ordinary-gradient discretization with $A^\sharp$ has a strictly larger local stability interval,

$$0 < \gamma < \frac{2}{\lambda_{\max}(K_\beta)},$$

and a strictly larger no-sign-flip interval than the vanilla parameter gradient. In particular, there exists a nonempty range of γ for which the stiffest vanilla mode flips sign while no $A^\sharp$ mode does.

What does *not* survive automatically is condition-number dominance. Congruence by B can introduce its own ill-conditioning, so $\kappa(K_\beta) < \kappa(K)$ is not guaranteed for arbitrary networks. A useful bound is

$$\kappa(K_\beta) \le \kappa(B^\top B)\,\kappa(H_\beta), \qquad \kappa(K) \le \kappa(B^\top B)\,\kappa(H). \tag{48}$$

Finite lookahead improves the policy/objective factor $\kappa(H_\beta)$ but cannot by itself remove neural parameterization conditioning.

This distinction defines the intended theoretical scope: *the secant advantage survives as a better finite-target signal under ordinary gradients, while invariant shortest-path guarantees require additional assumptions on parameterization.*

7.10 Practical re-anchored training loop and staleness

A practical implementation is:

1. set the current actor π_k as both behavior and reference;
2. collect on-policy rollouts;

3. estimate $A^\sharp_{\pi_k,\beta}$, treating it as stop-gradient during the actor phase;

4. take one or a small number of PPO-like/SGD actor steps using $A^\sharp$ in place of A^π;

5. set the updated actor to π_{k+1} and immediately collect fresh data.

With one infinitesimal update per refresh, this is an Euler discretization of the parameter-space field $\mathcal{B}^*_\theta Y_\beta$. Multiple minibatch epochs keep using a stale secant anchored at π_k while the actor moves away; this is a finite-integration approximation and should be measured explicitly as a staleness error rather than silently identified with the ideal re-anchored field.

Two scales should be separated:

- $\beta = 1/\eta$ controls **lookahead distance**: how far ahead the direction generator queries the proximal solution;
- the optimizer learning rate / PPO clipping radius controls **actual movement**: how far the actor moves before re-anchoring.

This "look far, move little" separation is central to the plug-in interpretation.

8 Statistical Estimation, Credit Assignment, and Structural Limits

8.1 Critic parameterizations

For general bounded deterministic rewards, one may learn either a conditional return distribution $Z_\phi(G \mid s, a)$ and compute the log-MGF, or directly regress

$$h_\pi(s, a) = \mathbb{E}_\pi[e^{\beta G_t} \mid s, a]$$

(possibly in log-space). For binary RLVR, Eq. (62) shows that a calibrated success-probability critic is sufficient. This setting is especially useful for isolating optimization geometry from representational complexity.

State normalization

$$h_\pi(s) = \mathbb{E}_{a\sim\pi(\cdot|s)} h_\pi(s, a)$$

can be estimated by a state head, top-K action evaluation plus tail approximation, Monte Carlo action samples, or a joint architecture predicting state and action offsets. Normalization error directly perturbs $A^\sharp$ and should be separately measured.

8.2 GAE-style multi-step estimation of $A^\sharp$

The preceding sections treat $A^\sharp$ as a population-level signal and ask how it should be *used* once available. This subsection is deliberately orthogonal: it does not change the definition of $A^\sharp$, the proximal target, or any actor objective above. It only asks how to *estimate* $A^\sharp$ more effectively from finite on-policy rollouts. The motivation parallels generalized advantage estimation (GAE) [4]: rather than relying exclusively on a one-step bootstrap or exclusively on a full Monte Carlo return, combine multi-step rollout information with a learned value function to trade bootstrap error against sampling variance. Recent distributional advantage estimators also explore multi-step constructions, but the transformation below is specific to the log-MGF functional required by our finite-step proximal target [5].

Work in the exponential-value domain. Let $\beta = 1/\eta$ and define the transformed state and state-action values

$$\Psi_\pi(s_t) := \exp\Big(\beta V^\sharp_{\pi,\eta}(s_t)\Big) = \mathbb{E}_\pi\Big[e^{\beta G_t} \mid s_t\Big], \tag{49}$$

$$\Psi_\pi(s_t, a_t) := \exp\Big(\beta Q^\sharp_{\pi,\eta}(s_t, a_t)\Big) = \mathbb{E}_\pi\Big[e^{\beta G_t} \mid s_t, a_t\Big]. \tag{50}$$

Although $V^\sharp$ itself is a nonlinear log-MGF, Ψ_π is again an ordinary conditional expectation. Under the deterministic autoregressive dynamics assumed throughout the proposal,

$$\Psi_\pi(s_t, a_t) = e^{\beta r_t}\Psi_\pi(s_{t+1}), \tag{51}$$

$$\Psi_\pi(s_t) = \mathbb{E}_{a_t\sim\pi(\cdot|s_t)}\left[\Psi_\pi(s_t, a_t)\right]. \tag{52}$$

Moreover,

$$\boxed{e^{\beta A^\sharp_{\pi,\eta}(s,a)} = \frac{\Psi_\pi(s,a)}{\Psi_\pi(s)}.} \tag{53}$$

Thus the natural quantity for multi-step estimation is Ψ, or equivalently the exact target ratio $e^{\beta A^\sharp}$, rather than a naive additive TD residual in $V^\sharp$-space.

n-step transformed returns. Suppose a rollout fixes the sampled action a_t and then follows the current reference policy π thereafter. For any bootstrap depth $n \geq 1$, define

$$Y_t^{(n)} := \exp\left(\beta\sum_{j=0}^{n-1} r_{t+j}\right)\widehat{\Psi}_\phi(s_{t+n}), \tag{54}$$

with the obvious terminal convention when the episode ends before $t+n$. If the bootstrap critic is exact, $\widehat{\Psi}_\phi = \Psi_\pi$, the tower property gives

$$\boxed{\mathbb{E}_\pi\Big[Y_t^{(n)} \mid s_t, a_t\Big] = \Psi_\pi(s_t, a_t) \qquad \text{for every } n.} \tag{55}$$

Equation (55) is the direct analogue of the fact that ordinary n-step bootstrapped returns share the same conditional expectation when the conventional value function is exact.

An entropic λ-return. Let N_t denote the remaining rollout horizon. A GAE-like mixture of all available bootstrap depths is

$$Y_t^{(\lambda)} := (1-\lambda)\sum_{n=1}^{N_t-1}\lambda^{n-1}Y_t^{(n)} + \lambda^{N_t-1}Y_t^{(N_t)}, \qquad \lambda \in [0,1]. \tag{56}$$

With an exact bootstrap critic, the weights sum to one and therefore

$$\mathbb{E}_\pi\Big[Y_t^{(\lambda)} \mid s_t, a_t\Big] = \Psi_\pi(s_t, a_t). \tag{57}$$

The same estimator admits a backward recursion analogous to TD(λ):

$$\boxed{Y_t^{(\lambda)} = e^{\beta r_t}\left[(1-\lambda)\widehat{\Psi}_\phi(s_{t+1}) + \lambda Y_{t+1}^{(\lambda)}\right].} \tag{58}$$

Equivalently, defining the transformed TD residual

$$\delta_t^{\Psi} := e^{\beta r_t}\widehat{\Psi}_\phi(s_{t+1}) - \widehat{\Psi}_\phi(s_t),$$

one obtains a multiplicatively weighted analogue of the usual GAE residual expansion. The propagation factor is no longer a fixed discount alone; it also contains the accumulated exponential reward along the intervening rollout segment.

How the estimator interfaces with $A^\sharp$. If the state normalizer $\Psi_\pi(s_t)$ were exact, then

$$\widehat{w}_t^{(\lambda)} := \frac{Y_t^{(\lambda)}}{\Psi_\pi(s_t)} \tag{59}$$

is an unbiased estimator of the exact proximal policy ratio

$$w_t^\star = e^{\beta A^\sharp_{\pi,\eta}(s_t,a_t)} = \frac{q^\star_{\pi,\eta}(a_t \mid s_t)}{\pi(a_t \mid s_t)}.$$

However, unbiasedness in ratio space does *not* survive the logarithm:

$$\mathbb{E}[\widehat{w}_t^{(\lambda)}] = w_t^\star \quad \not\Rightarrow \quad \mathbb{E}\Big[\eta \log \widehat{w}_t^{(\lambda)}\Big] = A^\sharp_{\pi,\eta}(s_t,a_t).$$

Indeed, Jensen's inequality generally produces a downward log bias. Therefore the recommended reverse-KL implementation is *not* to take the logarithm of a single noisy λ-return and call it an unbiased $A^\sharp$ sample. Instead, use $Y_t^{(\lambda)}$ as a multi-step target for learning a positive critic Ψ_ϕ (or an equivalent log-critic), and then form the plug-in estimate

$$\widehat{A}^\sharp_\phi(s,a) := \eta \log \frac{\widehat{\Psi}_\phi(s,a)}{\widehat{\Psi}_\phi(s)}. \tag{60}$$

This estimated $A^\sharp$ is then used, with stop-gradient through the critic, in whichever actor construction was selected in Sections 5–7. Consequently, the λ-return machinery changes only the statistical estimator of the finite-step signal; it leaves all preceding population-level definitions, surrogate identities, geometric theorems, and algorithmic uses of $A^\sharp$ unchanged.

An optional alternative is forward-KL/weighted distillation. Because Eq. (59) targets $w_t^\star$ directly, an unbiased ratio estimator can be inserted directly into a weighted log-likelihood gradient without taking a logarithm. This may be statistically attractive, but it changes the actor projection geometry from the reverse-KL surrogate analyzed earlier and should therefore be treated as a separate practical variant rather than as a replacement for the main method.

Bias–variance role of λ. The interpretation of λ parallels ordinary GAE but in the transformed domain. At $\lambda = 0$, Eq. (58) relies almost entirely on a one-step bootstrap and therefore has low rollout variance but can inherit critic bias. As $\lambda \to 1$, the target approaches the full Monte Carlo exponential return $e^{\beta G_t}$, reducing bootstrap dependence but potentially increasing variance sharply, especially for large β because rare high-return trajectories receive exponentially large weights. The practical role of λ is therefore to choose an estimation point on a bootstrap-bias versus exponential-return-variance continuum, independently of the policy-space role of β and independently of how aggressively the actor is actually updated.

Binary RLVR specialization. For $G \in \{0, 1\}$,

$$\Psi_\pi(s, a) = 1 + \big(e^\beta - 1\big)p_\pi(s, a), \qquad p_\pi(s, a) = \Pr_\pi(G = 1 \mid s, a). \tag{61}$$

Hence estimating Ψ is equivalent to estimating the scalar success probability. In this setting one may apply ordinary multi-step/TD(λ) machinery to the success-probability critic (or equivalently to its affine transform Ψ), and only afterwards apply the deterministic nonlinear mapping in Eq. (60). No expressive distributional critic is required. This specialization cleanly preserves the earlier Bernoulli-sufficiency observation: GAE-style estimation improves how the same sufficient statistic is estimated; it does not create additional return-distribution information.

8.3 Approximation–estimation trade-off

A mean-only update approximates the finite target using the first cumulant. If $\sigma_a^2 = \mathrm{Var}(G \mid s, a)$, then for large η the normalized target error satisfies

$$\log \frac{q^\star(a \mid s)}{q_{\mathrm{mean}}(a \mid s)} = \frac{\sigma_a^2 - \mathbb{E}_{a' \sim \pi}\sigma_{a'}^2}{2\eta^2} + O(\eta^{-3}),$$

and the corresponding one-hop target KL is $O(\eta^{-4})$. By contrast, critic error is amplified according to Eq. (25). The useful regime is therefore expected to be intermediate: finite enough that mean-only local targeting is measurably biased, but not so aggressive that exponential-moment estimation dominates.

The K-step and re-anchored methods offer two different responses. K-step optimization explicitly reduces per-hop β by splitting a fixed total displacement. Re-anchored plug-in optimization can keep a finite lookahead β while taking a much smaller actual actor step, but it still requires estimating the more aggressive lookahead signal.

8.4 Rao–Blackwellized token credit assignment

Sequence-level exponential reweighting can use

$$W = e^{\beta G}.$$

For a token score vector $g(s, a)$, the raw estimator is $X = Wg(s, a)$. Let

$$h(s, a) = \mathbb{E}[W \mid s, a], \qquad \widetilde{X} = h(s, a)g(s, a) = \mathbb{E}[X \mid s, a].$$

Candidate Theorem 8.1 (Rao–Blackwell variance reduction)**.** *For any square-integrable $g(s, a)$ measurable with respect to (s, a),*

$$\mathbb{E}[\widetilde{X}] = \mathbb{E}[X], \qquad \mathrm{Cov}(\widetilde{X}) \preceq \mathrm{Cov}(X).$$

This result compares two estimators of the same finite-step exponential target. It does not imply lower variance than vanilla policy gradient; the exponential transform itself can be statistically difficult.

8.5 Deterministic-dynamics limitation

The exact factorization relies structurally on deterministic transitions. In a stochastic environment, the same marginal return distribution can result from controllable future policy branching or uncontrollable transition randomness. A reference return distribution alone cannot generally distinguish these cases. Equivalently, Bellman recursion places the environment expectation outside the optimal value, while a naive log-MGF construction can exchange an environment expectation and a logarithm. Extending the theory to agentic RL with stochastic tools/users therefore requires an explicit model or Bellman treatment that separates policy-controlled branching from transition uncertainty.

9 Experimental Plan

9.1 Phase I: exact deterministic tree MDPs

The first phase should test the theory before introducing critic or neural approximation error. Construct finite deterministic trees where $\mathcal{T}_\beta$, $A^\sharp$, all occupancies, and exact natural gradients can be enumerated.

Core comparisons.

- Direct proximal policy-gradient flow versus the fixed-target $A^\sharp$ surrogate flow.
- One-step versus K-step exact and inexact surrogate optimization.
- Re-anchored tangent versus secant fields, including their Euler discretizations.
- The same methods after introducing an explicit finite-dimensional parameterization and ordinary Euclidean gradient updates.

Diagnostic constructions.

- Same mean, different continuation-return shape.
- Narrow-valley examples in pairwise log-odds coordinates, including single transverse excursions and repeated oscillation for graded return distributions.
- Controlled Hessian spectra to verify $H \mapsto H_\beta$ and the predicted stability/no-sign-flip ranges.
- Controlled critic noise and deliberately incomplete inner solves to test Eq. (31).

Metrics. Measure exact target KL, log-odds path length, transverse-motion energy, number of actor steps to a fixed target gap, spectral condition estimates, largest stable learning rate, sign-flip/oscillation frequency, and gradient covariance.

9.2 Phase II: binary RLVR

Use a compact open LLM (roughly 1.5B–4B parameters) on verifiable mathematical reasoning tasks. A binary success-probability critic isolates the optimization effect because no rich distribution model is needed.

Compare:

- PPO/GRPO with conventional advantages [1, 2];
- fixed-reference one-step $A^\sharp$ surrogate optimization;
- K-step $A^\sharp$ optimization for several K;
- re-anchored plug-in $A^\sharp$ with the same PPO/SGD optimizer as the conventional baseline;
- sequence-level PMD/Kimi-style exponential updates [6, 7, 8];
- categorical/distributional critic baselines that retain the conventional actor objective [9].

Primary outcomes should include pass@1, rollout/sample efficiency, training stability, KL moved per refresh, critic calibration, and target alignment. To isolate the plug-in effect, the most important ablation is A^π versus $A^\sharp$ under the *same* actor optimizer, data, clipping rule, and number of epochs.

9.3 Phase III: graded deterministic rewards

Binary RLVR cannot expose action-dependent higher moments beyond success probability. Use code generation scored by fraction of tests passed, multi-level verifier rewards, or other deterministic graded objectives. These tasks should test whether the variance/skewness corrections in Eq. (45) predict meaningful changes in actor directions.

9.4 Ablations aligned with the theory

- Lookahead β versus actual optimizer step size / PPO clip range.
- Number of K-step partitions at fixed total inverse temperature $\beta_\star$.
- Number of minibatch epochs before re-anchoring, measuring stale-direction error.
- Fixed-reference behavior sampling versus state-aggregation roll-in with frozen-reference continuation labels.
- Exact/reverse-KL surrogate versus forward-KL distillation versus plug-in PPO clipping.
- Raw sequence exponential weights versus conditionalized token weights.
- Critic-error injection and calibration.
- Local parameterization conditioning, e.g. low-dimensional softmax policies where B and F can be computed exactly.

10 Related Work

10.1 Trust regions, mirror descent, and exponential policy improvement

TRPO and PPO are canonical local trust-region policy-gradient methods [10, 1]. REPS derives relative-entropy policy-search updates [11]; MPO separates nonparametric improvement from parametric projection [12]; AWR and reward-weighted regression fit actors to return-weighted data [13, 14]. Policy mirror descent provides a general framework for KL/Bregman proximal policy optimization and connects to natural policy gradient [6, 7]. Lookahead PMD uses multi-step

environment lookahead [15]; our use of "lookahead" instead refers to a finite distance along the proximal/NPG policy-improvement flow.

Kimi k1.5 is especially relevant because it uses moving-reference relative-entropy optimization and a sequence-level exponential solution [8]. The proposal therefore does not claim novelty for moving-reference KL-proximal objectives. The intended contribution is the token-level finite-target advantage, its role as a direction generator, and the tangent-versus-secant optimization interpretation.

10.2 $Q\sharp$, twisted inference, and control as inference

$Q\sharp$ provides the closed-form deterministic-MDP log-MGF identity on which the proposal is built [3]. Twisted SMC and trust-region twisted policy improvement use related future-potential/Doob-h transforms [16, 17]. The distinction here is not the mathematical existence of an exponential twist, but using its conditional ratio as an online actor-learning signal and studying how much finite-target geometry survives practical parameter gradients.

10.3 Distributional RL and risk-sensitive objectives

Distributional RL models return distributions for value prediction, robustness, exploration, or distribution-aware advantages [18, 19, 5]. Recent LLM work includes DVPO, DFPO, and categorical critics [20, 21, 9]. Risk-sensitive RL also uses entropic log-MGF objectives [22, 23]. Our interpretation differs: the global reward objective need not be risk-sensitive; the log-MGF appears because it parameterizes the exact finite KL-proximal improvement target.

10.4 Planning and multi-step gradient information

Other work augments local policy gradients with nonlocal future information, for example tree-expanded policy gradients [24]. Such work supports the broad idea that local gradients can benefit from lookahead. The mechanism here is different: the lookahead is not an explicit multi-transition planning tree, but a finite displacement along an exact exponential policy-improvement semigroup.

11 Limitations and Non-Claims

- **Deterministic dynamics are structural.** The exact target factorization does not generally survive stochastic environment transitions without additional modeling.
- **Support cannot be created.** Exponential reweighting amplifies successful trajectories already in support but cannot discover zero-probability modes.
- **Binary rewards contain no extra distributional degrees of freedom.** Any gains there must come from finite-step geometry and credit assignment.
- **Fixed-target natural-gradient optimality is not an SGD theorem.** The exponential path is Pareto-optimal/shortest only in the specified nonparametric geometries. Arbitrary neural parameterization can distort it.
- **The secant field need not be the gradient of one global scalar potential.** It is an autonomous vector field constructed by re-anchoring local surrogate gradients. Integrability is not assumed.

- **Same stationary points are an ideal policy-space result.** In parameter space, $\mathcal{B}_\theta^* Y_\beta$ can vanish even when $Y_\beta \neq 0$ if the parameterization cannot express or detect the relevant direction.
- **Condition-number improvement does not automatically survive arbitrary parameterization.** What survives generally is target-alignment structure and, under the local linear model, a reduction in the largest pulled-back curvature/stability bottleneck.
- **K-step semigroup equivalence assumes exact inner solutions.** Inexact projection and changing data distributions can accumulate error, motivating the re-anchored plug-in variant.
- **Small η / large β can be statistically expensive.** The same finite lookahead that reduces local optimization myopia can amplify rare-return estimation error.

12 Discussion

12.1 Binary RLVR and what $A^\sharp$ does not add

If $G \in \{0, 1\}$ and

$$p_\pi(s, a) = \Pr_\pi(G = 1 \mid s, a), \qquad p_\pi(s) = \Pr_\pi(G = 1 \mid s),$$

then

$$A^\sharp_{\pi,\eta}(s, a) = \eta \log \frac{1 + (e^{1/\eta} - 1)p_\pi(s, a)}{1 + (e^{1/\eta} - 1)p_\pi(s)}. \tag{62}$$

Thus no expressive distributional model is necessary in strict binary RLVR: the return distribution is Bernoulli and fully described by its success probability. Action ranking agrees with the conventional advantage. The contribution in this regime is finite-step nonlinear geometry and token-level credit assignment, not richer distributional representation.

13 Research Questions

The restructured project is organized around the following questions.

1. Under exactly which deterministic autoregressive assumptions does the $Q\sharp$ finite-target identity support the surrogate and direction-generation constructions used here, and where does that interface fail?
2. How much shorter is the ideal fixed-target $A^\sharp$ path than direct proximal/NPG paths in controlled problems, and when is the difference substantial rather than cosmetic?
3. What is the optimal K in the path-length versus critic-estimation trade-off, and how severe is accumulated error when inner subproblems are not solved exactly?
4. Does the re-anchored secant field predict larger stable actor steps and reduced zig-zagging in exact and parameterized toy models?
5. Under ordinary neural parameter gradients, how much of the finite-target alignment in Theorem 7.6 survives in practice?
6. In binary RLVR, does plug-in $A^\sharp$ improve optimization despite using only success-probability information?

7. In graded deterministic rewards, do higher-cumulant corrections produce measurable directional gains beyond mean-only advantages?

8. How should lookahead β be scheduled relative to the actual optimizer step size, critic uncertainty, and rollout budget?

14 Milestones

1. **Formal theory pass.** Treat the $Q\sharp$ factorization as a cited foundation, then fully verify Proposition 4.1, Theorems 4.2–7.6, the K-step limit, and all local spectral statements; produce counterexamples where parameterization destroys stronger claims.

2. **Exact tree-MDP implementation.** Visualize tangent, fixed-target, piecewise-K, and re-anchored secant trajectories; verify path length, stability, and oscillation predictions.

3. **Parameterized toy models.** Use softmax neural parameterizations with controllable Jacobian conditioning to test Eqs. (43)–(48).

4. **Binary RLVR prototype.** Implement a calibrated success-probability critic and compare one-step, K-step, and plug-in $A^\sharp$ actor updates in an existing PPO/GRPO stack.

5. **Graded-reward extension.** Test whether return-shape corrections predicted by the cumulant expansion improve direction quality and sample efficiency.

15 Conclusion

The proposal begins from a standard KL-proximal policy-improvement problem and asks how best to exploit an unusual fact: in deterministic autoregressive RL, the finite optimum is analytically characterized. Direct policy gradient treats the proximal objective as any other smooth function and initially uses only a local tangent that is blind to the intended step size. $A^\sharp$ instead converts the exact finite target into a token-level log-policy displacement.

This signal supports three increasingly online uses. One-step surrogate optimization aims directly at the fixed analytic target and enjoys strong ideal information-geometric properties, but aggressive finite targets are statistically difficult. K-step decomposition reduces that estimation burden by replacing one large target with several easier ones, at the cost of a longer piecewise path and possible accumulated inner-solve error. Re-anchored plug-in optimization takes the most radical view: the proximal problem need never be solved. It can be queried at every current policy merely to provide a finite-lookahead direction. In the ideal policy space this yields a secant field—an average of future NPG tangents—with the same stationary points and favorable local conditioning. In a neural actor, ordinary gradients distort this geometry, but they still backpropagate the secant target rather than the tangent target, preserving exact first-order finite-target alignment and useful local stability structure.

The resulting research program is therefore not "distributional RL is richer than scalar RL." It is a more specific optimization hypothesis: *when a finite policy-improvement endpoint is available, distilling its relative location into the actor's current search direction can reduce the waste caused by purely local re-linearization, provided the statistical cost of estimating that lookahead remains manageable.*

A Proof Sketches and Derivations

A.1 Exact factorization

Condition Eq. (6) on a prefix state $s_t = s$ and next action $a_t = a$. Deterministic transitions make the trajectory likelihood ratio after fixing (s, a) equal to the continuation exponential moment, yielding

$$q^\star(a \mid s) = \pi(a \mid s)\frac{\mathbb{E}_\pi[e^{\beta G_t} \mid s, a]}{\mathbb{E}_\pi[e^{\beta G_t} \mid s]}.$$

Taking logs gives Eq. (11).

A.2 Surrogate equivalence

Using $A^\sharp = \eta \log(q^\star/\pi)$,

$$\begin{aligned}\mathbb{E}_q[A^\sharp] - \eta D_{\mathrm{KL}}(q\|\pi) &= \eta\mathbb{E}_q \log\frac{q^\star}{\pi} - \eta\mathbb{E}_q \log\frac{q}{\pi} \\ &= -\eta D_{\mathrm{KL}}(q\|q^\star).\end{aligned}$$

Average over μ.

A.3 Strong-KL expansion

For bounded G,

$$\log \mathbb{E}[e^{\beta G}] = \beta\kappa_1 + \frac{\beta^2}{2}\kappa_2 + \frac{\beta^3}{6}\kappa_3 + O(\beta^4).$$

Multiply by $1/\beta = \eta$ and subtract the state baseline from the state-action quantity.

A.4 Target-KL Pareto path

Fix any evaluation weighting μ_{eval}. The constrained problem uses the same μ_{eval} in the target divergence and the reference-KL constraint. Introducing one global multiplier $\lambda \geq 0$, the factors $\mu_{\mathrm{eval}}(s)$ cancel from the statewise first-order conditions, leaving, for every state in its support,

$$q(a \mid s) \propto q^\star(a \mid s)^{1/(1+\lambda)}\pi(a \mid s)^{\lambda/(1+\lambda)}.$$

Thus the optimizer is Eq. (17) with the same $\alpha = (1+\lambda)^{-1}$ at every state. This also makes clear why the theorem holds for any fixed μ_{eval} and does not require $\mu_{\mathrm{eval}} = \mu_{\mathrm{sur}}$. Monotonicity of the radius follows from standard exponential-family derivatives.

A.5 Frozen-surrogate natural-gradient flow

Write the surrogate weighting as μ_{sur} and equip the simplex with the Fisher metric weighted by the same measure. The factor $\mu_{\mathrm{sur}}(s)$ then appears in both the differential of the surrogate and the metric and cancels statewise. Consequently, for pairwise log-odds,

$$\frac{d}{dt}\left[\log\frac{q_t(a \mid s)}{q_t(b \mid s)} - \log\frac{q^\star(a \mid s)}{q^\star(b \mid s)}\right] = -\eta\left[\log\frac{q_t(a \mid s)}{q_t(b \mid s)} - \log\frac{q^\star(a \mid s)}{q^\star(b \mid s)}\right].$$

Hence every state contracts with the common factor $e^{-\eta t}$ and Eq. (19) follows. If the metric instead uses an equivalent measure μ_{met}, the right-hand side is multiplied by $d\mu_{\mathrm{sur}}/d\mu_{\mathrm{met}}(s)$; the common-$\alpha$ flow is therefore lost unless the two weightings are proportional.

A.6 Log-odds shortest path

Fix any ambient weighting $\mu_{\log}$ and full-support action measure ν. Centering removes state-wise normalization constants, so

$$z^{(\nu)}_{q_\alpha} = (1-\alpha) z^{(\nu)}_{\pi} + \alpha z^{(\nu)}_{q^\star}.$$

This identity is pointwise and is independent of $\mu_{\log}$. For any competing absolutely continuous path z_t in the corresponding Hilbert space,

$$\int_0^1 \|\dot z_t\|_{\mu_{\log},\nu}\, dt \geq \left\| \int_0^1 \dot z_t dt \right\|_{\mu_{\log},\nu} = \|z_1 - z_0\|_{\mu_{\log},\nu}\,.$$

The affine $A^\sharp$ path attains equality for every such fixed ambient weighting.

A.7 Semigroup and K-step limit

Two exponential tilts multiply:

$$\frac{Pe^{\beta_1 G}}{Z_1} \mapsto \frac{Pe^{(\beta_1+\beta_2)G}}{Z_{12}}.$$

This proves Eq. (26). Let $z(u) = z^{(\nu)}_{\mathcal{T}_u(\pi_0)}$ in a fixed log-odds Hilbert geometry and let $h = \beta_\star/K$. The semigroup identity gives exact knot points $z(u_j)$ with $u_j = jh$. Since the inner frozen-target exponential interpolation is affine in z, the geometrically parameterized K-hop path is exactly the piecewise-linear interpolant $\widetilde z_K$ of these knots.

Assume $z \in C^2([0,\beta_\star];\mathcal{H})$ and let $M = \sup_u \|z''(u)\| < \infty$. Standard Hilbert-valued interpolation estimates give

$$\sup_u \|\widetilde z_K(u) - z(u)\| = O(Mh^2) = O(K^{-2}),$$

while on each open mesh interval the secant slope is the average of z' over that interval, yielding

$$\operatorname*{ess\,sup}_u \left\| \dot{\widetilde z}_K(u) - \dot z(u) \right\| = O(Mh) = O(K^{-1}).$$

Finally, Section 2 identifies $u \mapsto \mathcal{T}_u(\pi_0)$ with the ordinary nonparametric NPG flow, hence $\dot z(u) = X_{\mathrm{NPG}}(z(u))$. This proves Eqs. (28)–(29) and specifies the precise sense in which the piecewise finite-target construction approaches the local NPG path.

A.8 Hilbert-metric drift bound

For any P, Q,

$$\log \frac{\mathcal{T}_\beta(P)}{\mathcal{T}_\beta(Q)} = \log \frac{P}{Q} + C(P,Q),$$

where C is trajectory-independent, so the essential range of the log-ratio is unchanged and $\mathcal{T}_\beta$ is an isometry in d_H. Repeated triangle inequalities yield Eq. (31).

A.9 Secant representation

By the fundamental theorem of calculus along the semigroup orbit,

$$z(\Phi_\beta(z)) - z = \int_0^\beta \frac{d}{du} z(\Phi_u(z)) du = \int_0^\beta X(\Phi_u(z)) du.$$

Divide by β. Taylor-expanding $X(\Phi_u(z))$ around $u = 0$ yields Eq. (34).

A.10 Stationary points and contraction

If $Y_\beta(z) = 0$, then $\Phi_\beta(z) = z$. Monotonicity of J along its gradient flow gives

$$0 = J(\Phi_\beta(z)) - J(z) = \int_0^\beta \|\mathrm{grad}_h\, J(\Phi_u(z))\|_h^2\, du,$$

so the trajectory is stationary. For the contraction result,

$$\begin{aligned}\frac{1}{2}\frac{d}{dt}\|z - z^\star\|^2 &= \frac{1}{\beta}\langle z - z^\star, \Phi_\beta(z) - z\rangle \\ &\le \frac{q_\beta - 1}{\beta}\|z - z^\star\|^2.\end{aligned}$$

Apply Grönwall's inequality.

A.11 Spectral damping and discrete stability

For $\dot e = -He$, $\Phi_\beta(e) = e^{-\beta H}e$, so Eq. (37) follows. Functional calculus gives eigenvalues $\phi_\beta(\lambda)$. Since $(1 - e^{-x})/x$ decreases for $x > 0$, the condition-number inequality follows. Euler stability of an SPD linear system requires $|1 - \gamma\lambda_i| < 1$ for every mode; absence of sign flips requires $1 - \gamma\lambda_i \ge 0$.

A.12 Parameter-space target alignment

Linearize the log policy:

$$\delta \log \pi = \mathcal{B}_\theta v.$$

Then

$$\langle \mathcal{B}_\theta v, \Delta_\beta\rangle = \langle v, \mathcal{B}_\theta^* \Delta_\beta\rangle.$$

Cauchy–Schwarz under $\|v\|_2 \le c$ gives Theorem 7.6. Since $\Delta_\beta = \beta A^\sharp$, the maximizing direction is the ordinary plug-in $A^\sharp$ policy gradient.

A.13 Local parameter-space stability

Under $z - z^\star = B(\theta - \theta^\star)$, pulling back the quadratic tangent and secant fields yields $K = B^\top H B$ and $K_\beta = B^\top H_\beta B$. The Loewner inequality $0 \prec H_\beta \prec H$ implies $0 \prec K_\beta \prec K$ for full-column-rank B, hence the maximum-eigenvalue and stability claims. Bounds in Eq. (48) follow from standard Rayleigh quotient inequalities.

A.14 Rao–Blackwellization

With $X = Wg(s,a)$ and $\widetilde{X} = \mathbb{E}[X \mid s,a]$, the law of total covariance gives

$$\mathrm{Cov}(X) = \mathrm{Cov}(\widetilde{X}) + \mathbb{E}[\mathrm{Cov}(X \mid s,a)],$$

and the second term is positive semidefinite.

B Notation Summary

Symbol	Meaning
π	Current/reference policy at a generic local problem.
π_k	Current actor at online iteration k.
$q^\star_{\pi,\eta}$	Exact optimizer of the KL-proximal problem around π.
η	KL temperature; larger η means more conservative finite targets.
$\beta = 1/\eta$	Inverse temperature / lookahead distance used by the semigroup.
G_t	Return from time/token t onward.
$Q^\sharp, V^\sharp, A^\sharp$	Conditional log-MGF state-action value, state value, and finite-step advantage.
$\mathcal{T}_\beta = \Phi_\beta$	Exact exponential-tilt / proximal semigroup operator.
X	Ordinary NPG tangent field in log-policy coordinates.
Y_β	Re-anchored finite-lookahead secant field.
μ	Frozen state weighting used for conditional-policy objectives.
ν	Fixed full-support action weighting used to define centered log-odds geometry.
$\mathcal{B}_\theta$	Score/Jacobian operator mapping parameter motion to infinitesimal log-policy motion.
$F_\theta = \mathcal{B}_\theta^* \mathcal{B}_\theta$	Fisher matrix under the chosen state/action weighting.
H, H_β	Local policy-space tangent and secant linearization matrices.
K, K_β	Their local Euclidean parameter-space pullbacks.

References

[1] John Schulman et al. "Proximal Policy Optimization Algorithms". In: *arXiv preprint arXiv:1707.06347* (2017).

[2] Zhihong Shao et al. "DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models". In: *arXiv preprint arXiv:2402.03300* (2024).

[3] Jin Peng Zhou et al. "Q♯: Provably Optimal Distributional RL for LLM Post-Training". In: *Advances in Neural Information Processing Systems*. Vol. 38. 2025.

[4] John Schulman et al. "High-Dimensional Continuous Control Using Generalized Advantage Estimation". In: *arXiv preprint arXiv:1506.02438* (2015).

[5] Shahil Shaik, Jonathon M. Smereka, and Yue Wang. "Generalized Advantage Estimation for Distributional Policy Gradients". In: *arXiv preprint arXiv:2507.17530* (2025).

[6] Guanghui Lan. "Policy Mirror Descent for Reinforcement Learning: Linear Convergence, New Sampling Complexity, and Generalized Problem Classes". In: *Mathematical Programming* 198.1 (2023), pp. 1059–1106. DOI: 10.1007/s10107-022-01816-5.

[7] Manan Tomar et al. "Mirror Descent Policy Optimization". In: *arXiv preprint arXiv:2005.09814* (2020).

[8] Kimi Team et al. "Kimi k1.5: Scaling Reinforcement Learning with LLMs". In: *arXiv preprint arXiv:2501.12599* (2025).

[9] Zhijian Zhou et al. "Start Classifying: Categorical Critics for LLM Reinforcement Learning". In: *arXiv preprint arXiv:2608.02181* (2026).

[10] John Schulman et al. “Trust Region Policy Optimization”. In: *arXiv preprint arXiv:1502.05477* (2015).

[11] Jan Peters, Katharina Mülling, and Yasemin Altun. “Relative Entropy Policy Search”. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 24. 1. 2010, pp. 1607–1612. DOI: 10.1609/aaai.v24i1.7727.

[12] Abbas Abdolmaleki et al. “Maximum a Posteriori Policy Optimisation”. In: *International Conference on Learning Representations*. 2018.

[13] Xue Bin Peng et al. “Advantage-Weighted Regression: Simple and Scalable Off-Policy Reinforcement Learning”. In: *arXiv preprint arXiv:1910.00177* (2019).

[14] Miroslav Štrupl et al. “Reward-Weighted Regression Converges to a Global Optimum”. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 36. 8. 2022, pp. 8361–8369. DOI: 10.1609/aaai.v36i8.20811.

[15] Kimon Protopapas and Anas Barakat. “Policy Mirror Descent with Lookahead”. In: *arXiv preprint arXiv:2403.14156* (2024).

[16] Stephen Zhao et al. “Probabilistic Inference in Language Models via Twisted Sequential Monte Carlo”. In: *arXiv preprint arXiv:2404.17546* (2024).

[17] Joery A. De Vries et al. “Trust-Region Twisted Policy Improvement”. In: *Proceedings of the 42nd International Conference on Machine Learning*. Vol. 267. Proceedings of Machine Learning Research. 2025, pp. 12901–12923.

[18] Marc G. Bellemare, Will Dabney, and Rémi Munos. “A Distributional Perspective on Reinforcement Learning”. In: *Proceedings of the 34th International Conference on Machine Learning*. Vol. 70. Proceedings of Machine Learning Research. 2017, pp. 449–458.

[19] Borislav Mavrin et al. “Distributional Reinforcement Learning for Efficient Exploration”. In: *Proceedings of the 36th International Conference on Machine Learning*. Vol. 97. Proceedings of Machine Learning Research. 2019, pp. 4424–4434.

[20] Dingwei Zhu et al. “DVPO: Distributional Value Modeling-based Policy Optimization for LLM Post-Training”. In: *arXiv preprint arXiv:2512.03847* (2025).

[21] Dingwei Zhu et al. “DFPO: Scaling Value Modeling via Distributional Flow towards Robust and Generalizable LLM Post-Training”. In: *arXiv preprint arXiv:2602.05890* (2026).

[22] David Nass, Boris Belousov, and Jan Peters. “Entropic Risk Measure in Policy Search”. In: *arXiv preprint arXiv:1906.09090* (2019).

[23] Yuhua Jiang et al. “Risk-Sensitive RL for Alleviating Exploration Dilemmas in Large Language Models”. In: *International Conference on Learning Representations*. 2026.

[24] Gal Dalal et al. “Policy Gradient with Tree Expansion”. In: *Proceedings of the 42nd International Conference on Machine Learning*. Vol. 267. Proceedings of Machine Learning Research. 2025, pp. 12229–12255.